## Mathematical and Physical Models

**Faraday's Law**

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt}\int_S \mathbf{B}\cdot d\mathbf{s} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}}$$

**Transformer**

$$V_{\text{emf}}^{\text{tr}} = -N\int_S \frac{\partial \mathbf{B}}{\partial t}\cdot d\mathbf{s} \qquad (N \text{ loops})$$

**Motional**

$$V_{\text{emf}}^{\text{m}} = \oint_C (\mathbf{u}\times\mathbf{B})\cdot d\mathbf{l}$$

**Charge–Current Continuity**

$$\nabla\cdot\mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

**EM Potentials**

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}$$
$$\mathbf{B} = \nabla\times\mathbf{A}$$

**Current Density**

Conduction $\mathbf{J}_c = \sigma\mathbf{E}$

Displacement $\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$

**Conductor Charge Dissipation**

$$\rho_v(t) = \rho_{vo}e^{-(\sigma/\varepsilon)t} = \rho_{vo}e^{-t/\tau_r}$$

## Important Terms

Provide definitions or explain the meaning of the following terms:

charge continuity equation
charge dissipation
displacement current $I_d$
electromagnetic induction
electromotive force $V_{\text{emf}}$
Faraday's law
Kirchhoff's current law
Lenz's law
motional emf $V_{\text{emf}}^{\text{m}}$
relaxation time constant
retarded potential
transformer emf $V_{\text{emf}}^{\text{tr}}$
wavenumber $k$

## PROBLEMS

### Sections 6-1 to 6-6: Faraday's Law and its Applications

*6.1 The switch in the bottom loop of Fig. P6.1 is closed at $t = 0$ and then opened at a later time $t_1$. What is the direction of the current $I$ in the top loop (clockwise or counterclockwise) at each of these two times?

6.2 The loop in Fig. P6.2 is in the $x$–$y$ plane and $\mathbf{B} = \hat{\mathbf{z}}B_0 \sin\omega t$ with $B_0$ positive. What is the direction of $I$ ($\hat{\boldsymbol{\phi}}$ or $-\hat{\boldsymbol{\phi}}$) at:

(a) $t = 0$
(b) $\omega t = \pi/4$
(c) $\omega t = \pi/2$

6.3 A coil consists of 100 turns of wire wrapped around a square frame of sides 0.25 m. The coil is centered at the origin with each of its sides parallel to the $x$ or $y$ axis. Find the induced emf across the open-circuited ends of the coil if the magnetic field is given by

*(a) $\mathbf{B} = \hat{\mathbf{z}}20e^{-3t}$ (T)
(b) $\mathbf{B} = \hat{\mathbf{z}}20\cos x \cos 10^3 t$ (T)
(c) $\mathbf{B} = \hat{\mathbf{z}}20\cos x \sin 2y \cos 10^3 t$ (T)

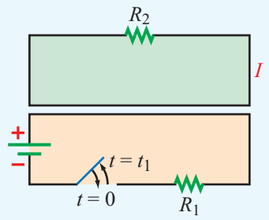



**Figure P6.1** Loops of Problem 6.1.

*Answer(s) available in Appendix E.

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## 数学与物理模型

**法拉第定律**

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt}\int_S \mathbf{B}\cdot d\mathbf{s} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}}$$

感生电动势

$$V_{\text{emf}}^{\text{tr}} = -N\int_S \frac{\partial \mathbf{B}}{\partial t}\cdot d\mathbf{s} \qquad (\text{N 匝})$$

动生电动势

$$V_{\text{emf}}^{\text{m}} = \oint_C (\mathbf{u}\times\mathbf{B})\cdot d\mathbf{l}$$

**电荷连续性方程**

$$\nabla\cdot\mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

**电磁势**

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}$$
$$\mathbf{B} = \nabla\times\mathbf{A}$$

**电流密度**

传导 $\mathbf{J}_c = \sigma\mathbf{E}$

位移 $\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$

**导体中的电荷弛豫**

$$\rho_v(t) = \rho_{vo}e^{-(\sigma/\varepsilon)t} = \rho_{vo}e^{-t/\tau_r}$$

## 重要术语

请给出下列术语的定义或解释其含义。

电荷连续性方程
电荷弛豫
位移电流 $I_d$
电磁感应
电动势 $V_{\text{emf}}$
法拉第定律
基尔霍夫电流定律
楞次定律
动生电动势
弛豫时间常数
推迟势
感生电动势
波数 k

## 问题

### 第 6-1 节至 6-6 节：法拉第定律及其应用

*6.1 图 P6.1 底部循环中的开关在 t = 0 处闭合，然后在稍后时间 $t_1$ 处打开。这两次时，顶部回路中的电流 I 的方向（顺时针或逆时针）是什么？

6.2 图 P6.2 中的环位于 x-y 平面中，且 $B = \hat{z}B_0 \sin \omega t$ 且 $B_0$ 为正。 I ($\hat{\boldsymbol{\phi}}$ 或 $-\hat{\boldsymbol{\phi}}$) 的方向是什么：

(a) t=0
(b) ωt=π/4
(c) ωt=π/2

6.3 边长0.25 m的正方形线圈有100匝，以原点为中心，各边平行于 x 或 y 轴。求右栏所列各磁场下线圈开路两端的感应电动势。

* 标记的习题在附录 E 提供答案。

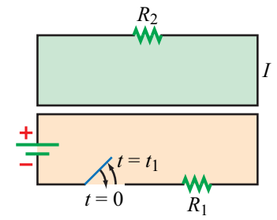



图 P6.1 习题6.1的两个回路。

习题6.3的磁通密度分别为：

*(a) $\mathbf{B} = \hat{\mathbf{z}}20e^{-3t}$ (T)
(b) $\mathbf{B} = \hat{\mathbf{z}}20\cos x \cos 10^3 t$ (T)
(c) $\mathbf{B} = \hat{\mathbf{z}}20\cos x \sin 2y \cos 10^3 t$ (T)

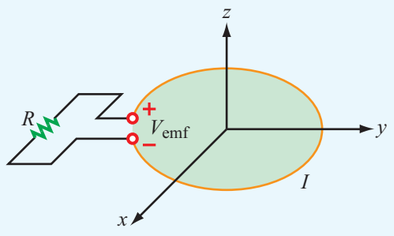



**Figure P6.2** Loop of Problem 6.2.

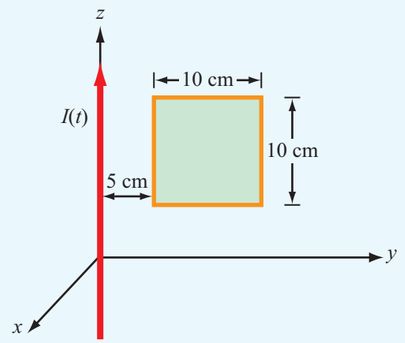



**Figure P6.6** Loop coplanar with long wire (Problem 6.6).

**6.4** A stationary conducting loop with an internal resistance of 0.5 Ω is placed in a time-varying magnetic field. When the loop is closed, a current of 5 A flows through it. What will the current be if the loop is opened to create a small gap and a 2 Ω resistor is connected across its open ends?

*6.5 A circular-loop TV antenna with 0.02 $m^2$ area is in the presence of a uniform-amplitude 300 Mhz signal. When oriented for maximum response, the loop develops an emf with a peak value of 30 (mV). What is the peak magnitude of **B** of the incident wave?

**6.6** The square loop shown in **Fig. P6.6** is coplanar with a long, straight wire carrying a current

$$I(t) = 5\cos(2\pi \times 10^4 t) \qquad \text{(A)}.$$

**(a)** Determine the emf induced across a small gap created in the loop.

**(b)** Determine the direction and magnitude of the current that would flow through a 4 Ω resistor connected across the gap. The loop has an internal resistance of 1 Ω.

*6.7 The rectangular conducting loop shown in **Fig. P6.7** rotates at 6,000 revolutions per minute in a uniform magnetic flux density given by

$$\mathbf{B} = \hat{y}\,50 \qquad \text{(mT)}.$$

Determine the current induced in the loop if its internal resistance is 0.5 Ω.

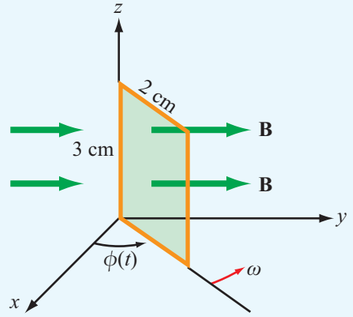



**Figure P6.7** Rotating loop in a magnetic field (Problem 6.7).

**6.8** The transformer shown in **Fig. P6.8** consists of a long wire coincident with the $z$ axis carrying a current $I = I_0 \cos \omega t$, coupling magnetic energy to a toroidal coil situated in the $x$–$y$ plane and centered at the origin. The toroidal core uses iron material with relative permeability $\mu_r$, around which 100 turns of a tightly wound coil serves to induce a voltage $V_{emf}$, as shown in the figure.

**(a)** Develop an expression for $V_{emf}$.

**(b)** Calculate $V_{emf}$ for $f = 60$ Hz, $\mu_r = 4000$, $a = 5$ cm, $b = 6$ cm, $c = 2$ cm, and $I_0 = 50$ A.

**6.9** A rectangular conducting loop 5 cm × 10 cm with a small air gap in one of its sides is spinning at 7200 revolutions per minute. If the field **B** is normal to the loop axis and its





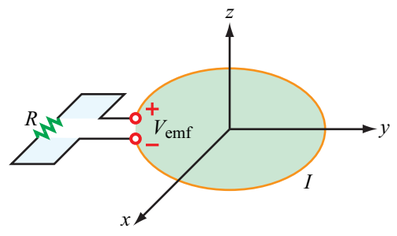



图 P6.2 问题 6.2 的循环。

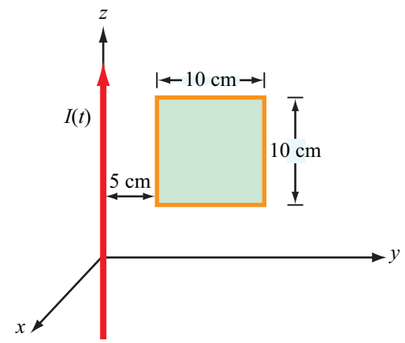



图 P6.6 环路与长导线共面（问题 6.6）。

6.4 将内阻为 0.5 Ω 的静止导电环置于时变磁场中。当环路闭合时，有 5 A 的电流流过。如果环路打开形成一个小间隙，并且在其开路端连接一个 2 Ω 电阻器，电流会是多少？

*6.5 面积为 0.02 $m^2$ 的圆环电视天线存在均匀幅度 300 M hz 信号。当面向最大响应时，环路产生峰值为 30 (mV) 的电动势。入射波 B 的峰值幅度是多少？

6.6 图 P6.6 所示的方形环路与承载电流的长直导线共面

$$I(t) = 5\cos(2\pi \times 10^4 t) \qquad \text{(A)}。$$

(a) 确定环路中产生的小间隙上感应的电动势。 (b) 确定流过连接在间隙上的 4 Ω 电阻的电流方向和大小。环路的内阻为 1 Ω。

*6.7 图 P6.7 所示的矩形导电环以每分钟 6,000 转的速度旋转，磁通密度由下式给出

$$\mathbf{B} = \hat{y}\,50 \qquad \text{(mT)}.$$

如果环路的内阻为 0.5 Ω，请确定环路中感应的电流。

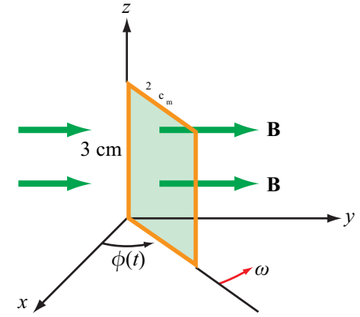



图 P6.7 磁场中的旋转环（问题 6.7）。

6.8 图 P6.8 所示的变压器由一根与 z 轴重合的长导线组成，承载电流 $I = I_0 \cos \omega t$，将磁能耦合到位于 x-y 平面并以原点为中心的环形线圈。环形磁芯采用相对磁导率 $\mu_r$ 的铁材料，在其周围紧密缠绕 100 匝线圈，用于感应电压 $V_{emf}$，如图所示。

(a) 开发 $V_{emf}$ 的表达式。 (b) 计算 $f = 60$ Hz、$\mu_r = 4000$、$a = 5$ cm、$b = 6$ cm、$c = 2$ cm 和 $I_0 = 50$ A 的 $V_{emf}$。

6.9 一个 5 cm × 10 cm 的矩形导电环，其一侧有一个小气隙，以每分钟 7200 转的速度旋转。如果场 B 垂直于循环轴且其

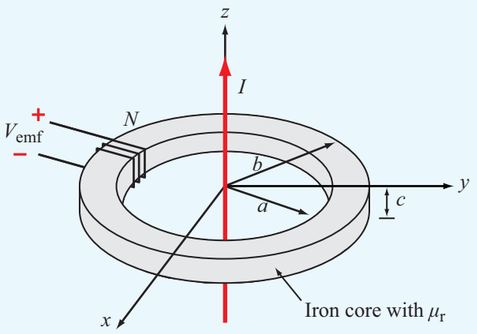



**Figure P6.8** Problem 6.8.

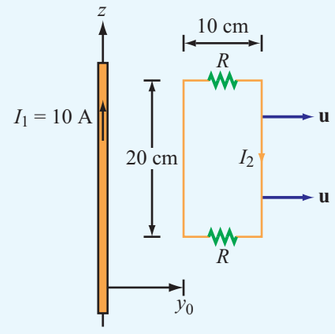



**Figure P6.11** Moving loop of Problem 6.11.

magnitude is $6 \times 10^{-6}$ T, what is the peak voltage induced across the air gap?

*6.10 A 50-cm long metal rod rotates about the $z$ axis at 90 revolutions per minute with end 1 fixed at the origin as shown in Fig. P6.10. Determine the induced emf $V_{12}$ if $\mathbf{B} = \hat{\mathbf{z}} 2 \times 10^{-4}$ T.

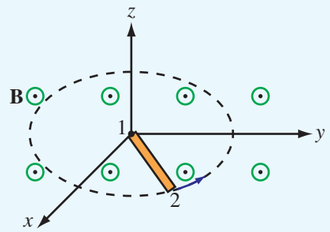



**Figure P6.10** Rotating rod of Problem 6.10.

6.11 The loop shown in Fig. P6.11 moves away from a wire carrying a current $I_1 = 10$ A at a constant velocity $\mathbf{u} = \hat{\mathbf{y}} 7.5$ (m/s). If $R = 10\ \Omega$ and the direction of $I_2$ is as defined in the figure, find $I_2$ as a function of $y_0$, which is the distance between the wire and the loop. Ignore the internal resistance of the loop.

*6.12 The electromagnetic generator shown in Fig. 6-12 is connected to an electric bulb with a resistance of 150 Ω. If the loop area is 0.1 $m^2$ and it rotates at 3,600 revolutions per minute in a uniform magnetic flux density $B_0 = 0.4$ T, determine the amplitude of the current generated in the light bulb.

6.13 The circular, conducting disk shown in Fig. P6.13 lies in the $x$–$y$ plane and rotates with uniform angular velocity $\omega$ about the $z$ axis. The disk is of radius $a$ and is present in a uniform magnetic flux density $\mathbf{B} = \hat{\mathbf{z}} B_0$. Obtain an expression for the emf induced at the rim relative to the center of the disk.

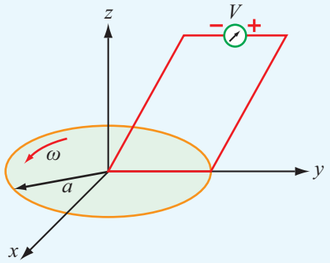



**Figure P6.13** Rotating circular disk in a magnetic field (Problem 6.13).

## Section 6-7: Displacement Current

6.14 The plates of a parallel-plate capacitor have areas of 10 $cm^2$ each and are separated by 2 cm. The capacitor is filled





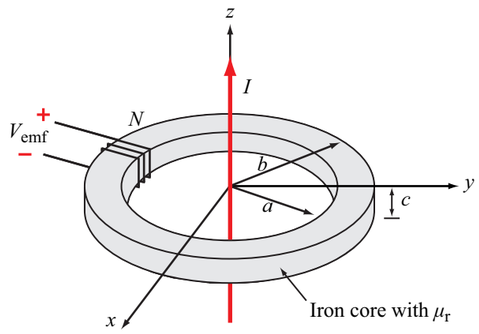



图 P6.8 问题 6.8。

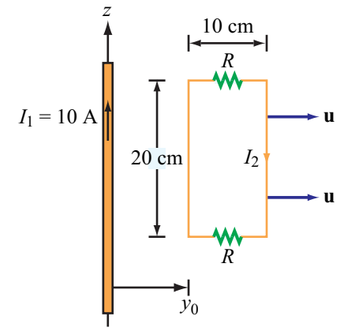



图 P6.11 问题 6.11 的移动循环。

幅度为 $6 \times 10^{-6}$ T，气隙上感应的峰值电压是多少？

*6.10 一根 50 厘米长的金属棒绕 z 轴以每分钟 90 转的速度旋转，一端固定在原点，如图 P6.10 所示。如果 $\mathbf{B} = \hat{\mathbf{z}} 2 \times 10^{-4}$ T，则确定感应电动势 $V_{12}$。

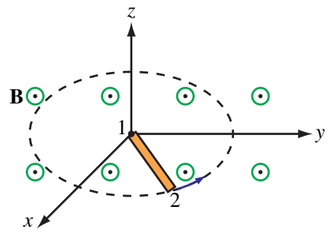



图 P6.10 问题 6.10 的旋转杆。

6.11 图 P6.11 所示的回路远离载有电流 I 的导线$_1$ =10 A 恒速 u= ŷ7.5（米/秒）。如果R= 10 Ω和我的方向$_2$如图所示，发现我$_2$作为 y 的函数$_0$，这是电线和环路之间的距离。忽略环路的内阻。

*6.12 将图6-12所示的电磁发电机与电阻为150Ω的电灯泡相连。如果环路面积为 0.1 $m^2$ 且转速为 3,600 转

每分钟在均匀磁通密度 $B_0 = 0.4$ T 下，确定灯泡中产生的电流的幅度。

6.13 图 P6.13 所示的圆形导电盘位于 x-y 平面上，并绕 z 轴以均匀角速度 ω 旋转。圆盘的半径为 a，磁通密度均匀为 $\mathbf{B} = \hat{\mathbf{z}} B_0$。获得相对于圆盘中心在边缘处感应的电动势的表达式。

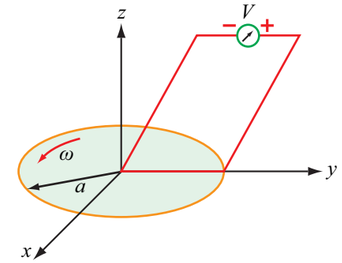



图 P6.13 在磁场中旋转圆盘（问题 6.13）。

第 6-7 节：位移电流

6.14 平行板电容器的极板面积为 10 $cm^2$，间距为 2 cm。电容已充满

with a dielectric material with $\varepsilon = 4\varepsilon_0$, and the voltage across it is given by $V(t) = 30\cos 2\pi \times 10^6 t$ (V). Find the displacement current.

*6.15 A coaxial capacitor of length $l = 6$ cm uses an insulating dielectric material with $\varepsilon_r = 9$. The radii of the cylindrical conductors are 0.5 cm and 1 cm. If the voltage applied across the capacitor is

$$V(t) = 50\sin(120\pi t) \qquad \text{(V)},$$

what is the displacement current?

6.16 The parallel-plate capacitor shown in Fig. P6.16 is filled with a lossy dielectric material of relative permittivity $\varepsilon_r$ and conductivity $\sigma$. The separation between the plates is $d$ and each plate is of area $A$. The capacitor is connected to a time-varying voltage source $V(t)$.

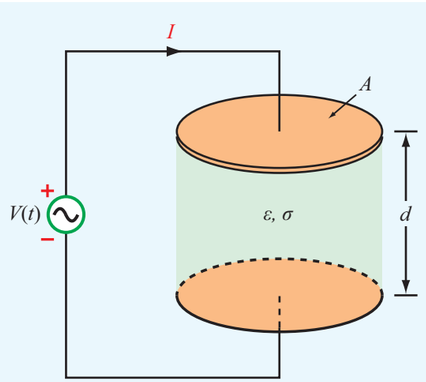



**Figure P6.16** Parallel-plate capacitor containing a lossy dielectric material (Problem 6.16).

(a) Obtain an expression for $I_c$, the conduction current flowing between the plates inside the capacitor, in terms of the given quantities.

(b) Obtain an expression for $I_d$, the displacement current flowing inside the capacitor.

(c) Based on your expressions for parts (a) and (b), give an equivalent-circuit representation for the capacitor.

(d) Evaluate the values of the circuit elements for $A = 4$ cm$^2$, $d = 0.5$ cm, $\varepsilon_r = 4$, $\sigma = 2.5$ (S/m), and

$$V(t) = 10\cos(3\pi \times 10^3 t) \quad \text{(V)}.$$

*6.17 In wet soil characterized by $\sigma = 10^{-2}$ (S/m), $\mu_r = 1$, and $\varepsilon_r = 36$, at what frequency is the conduction current density equal in magnitude to the displacement current density?

6.18 An electromagnetic wave propagating in seawater has an electric field with a time variation given by $\mathbf{E} = \hat{\mathbf{z}} E_0 \cos \omega t$. If the permittivity of water is $81\varepsilon_0$ and its conductivity is 4 (S/m), find the ratio of the magnitudes of the conduction current density to displacement current density at each of the following frequencies:

(a) 1 kHz

*(b) 1 MHz

(c) 1 GHz

(d) 100 GHz

## Sections 6-9 and 6-10: Continuity Equation and Charge Dissipation

6.19 At $t = 0$, charge density $\rho_{v0}$ was introduced into the interior of a material with a relative permittivity $\varepsilon_r = 9$. If at $t = 1$ $\mu$s the charge density has dissipated down to $10^{-3}\rho_{v0}$, what is the conductivity of the material?

*6.20 If the current density in a conducting medium is given by

$$\mathbf{J}(x,y,z;t) = (\hat{\mathbf{x}}z - \hat{\mathbf{y}}4y^2 + \hat{\mathbf{z}}2x)\cos \omega t,$$

determine the corresponding charge distribution $\rho_v(x,y,z;t)$.

6.21 In a certain medium, the direction of current density $\mathbf{J}$ points in the radial direction in cylindrical coordinates, and its magnitude is independent of both $\phi$ and $z$. Determine $\mathbf{J}$ given that the charge density in the medium is

$$\rho_v = \rho_0 r \cos \omega t \quad (\text{C/m}^3).$$

6.22 If we were to characterize how good a material is as an insulator by its resistance to dissipating charge, which of the following two materials is the better insulator?

| | | |
|---|---|---|
| Dry Soil: | $\varepsilon_r = 2.5$, | $\sigma = 10^{-4}$ (S/m) |
| Fresh Water: | $\varepsilon_r = 80$, | $\sigma = 10^{-3}$ (S/m) |

## Sections 6-11: Electromagnetic Potentials

6.23 The electric field of an electromagnetic wave propagating in air is given by

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}4\cos(6\times 10^8 t - 2z) + \hat{\mathbf{y}}3\sin(6\times 10^8 t - 2z) \qquad \text{(V/m)}.$$

Find the associated magnetic field $\mathbf{H}(z,t)$.





采用具有 ε = 4ε0 的介电材料，其两端电压由 V (t) = 30 c os2π × 10^6t (V) 给出。求位移电流。

*6.15 长度 l = 6 cm 的同轴电容器使用 εr = 9 的绝缘电介质材料。圆柱形导体的半径为 0.5 cm 和 1 cm。如果施加在电容器两端的电压为

$$V(t) = 50\sin(120\pi t) \qquad \text{(V)},$$

什么是位移电流?

6.16 图 P6.16 所示的平行板电容器填充有相对介电常数 εr 和电导率 σ 的有损介电材料。极板之间的间距为 d，每个极板的面积为 A。电容器连接到时变电压源 V (t)。

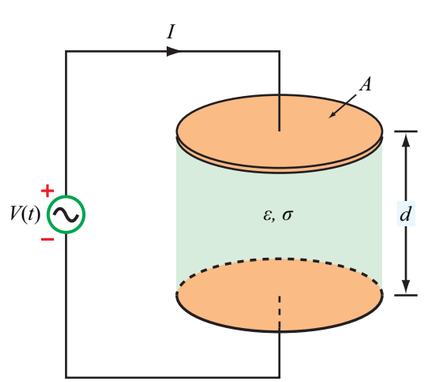



图 P6.16 包含有损耗介电材料的平行板电容器（问题 6.16）。

(a) 根据给定的量，获得 Ic 的表达式，即电容器内部极板之间流动的传导电流。

(b) 获得电容器内部流动的位移电流 Id 的表达式。

(c) 根据 (a) 和 (b) 部分的表达式，给出电容器的等效电路表示。

(d) 评估电路元件的值 A = 4 cm^2、d = 0.5 cm、εr = 4、σ = 2.5 (S/m) 和

$$V(t) = 10\cos(3\pi \times 10^3 t) \quad \text{(V)}.$$

*6.17 在以σ = 10^−2 (S/m)、μr = 1 和εr = 36 为特征的湿土壤中，在什么频率下传导电流密度的大小等于位移电流密度?

6.18 在海水中传播的电磁波具有随时间变化的电场，其变化由 E = ẑE0 cos ω t 给出。如果水的介电常数为 81ε0，电导率为 4 (S/m)，求以下每个频率下传导电流密度与位移电流密度的大小之比：

(a) 1 kHz *( b) 1 MHz (c) 1 GHz (d) 100 GHz

第 6-9 节和 6-10 节：连续性方程和电荷耗散

6.19 在 t = 0 处，电荷密度 ρv0 被引入相对介电常数为 εr = 的材料内部 9. 如果在 t = 1 μ s 处电荷密度已耗散至 10^−3ρv0，则该材料的电导率是多少?

*6.20 如果导电介质中的电流密度由下式给出

$$\mathbf{J}(x,y,z;t) = (\hat{\mathbf{x}}z - \hat{\mathbf{y}}4y^2 + \hat{\mathbf{z}}2x)\cos \omega t,$$

确定相应的电荷分布 ρv(x, y, z;t)。

6.21 在某种介质中，电流密度J的方向指向柱坐标中的径向，其大小与ϕ和z无关。鉴于介质中的电荷密度为，确定 J

$$\rho_v = \rho_0 r \cos \omega t \quad (\text{C/m}^3).$$

6.22 如果我们要通过材料对耗散电荷的抵抗力来表征材料作为绝缘体的好坏，那么以下两种材料中哪一种是更好的绝缘体?

| | | |
|---|---|---|
| Dry Soil: | $\varepsilon_r = 2.5$, | $\sigma = 10^{-4}$ (S/m) |
| Fresh Water: | $\varepsilon_r = 80$, | $\sigma = 10^{-3}$ (S/m) |

第 6-11 节：电磁势

6.23 在空气中传播的电磁波的电场由下式给出

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}4\cos(6\times 10^8 t - 2z) + \hat{\mathbf{y}}3\sin(6\times 10^8 t - 2z) \qquad \text{(V/m)}.$$

求出相关的磁场 H(z,t)。

*6.24 The magnetic field in a dielectric material with $\varepsilon = 4\varepsilon_0$, $\mu = \mu_0$, and $\sigma = 0$ is given by

$$\mathbf{H}(y,t) = \hat{\mathbf{x}}5\cos(2\pi \times 10^7 t + ky) \quad \text{(A/m)}.$$

Find $k$ and the associated electric field $\mathbf{E}$.

6.25 Given an electric field

$$\mathbf{E} = \hat{\mathbf{x}}E_0 \sin ay \cos(\omega t - kz),$$

where $E_0$, $a$, $\omega$, and $k$ are constants, find $\mathbf{H}$.

*6.26 The electric field radiated by a short dipole antenna is given in spherical coordinates by

$$\mathbf{E}(R,\theta;t) = \hat{\boldsymbol{\theta}}\,\frac{2\times 10^{-2}}{R}\sin\theta\,\cos(6\pi \times 10^8 t - 2\pi R) \quad \text{(V/m)}.$$

Find $\mathbf{H}(R,\theta;t)$.

6.27 A Hertzian dipole is a short conducting wire carrying an approximately constant current over its length $l$. If such a dipole is placed along the $z$ axis with its midpoint at the origin and if the current flowing through it is $i(t) = I_0 \cos \omega t$, find the following:

(a) The retarded vector potential $\widetilde{\mathbf{A}}(R,\theta,\phi)$ at an observation point $Q(R,\theta,\phi)$ in a spherical coordinate system.

(b) The magnetic field phasor $\widetilde{\mathbf{H}}(R,\theta,\phi)$.

Assume $l$ to be sufficiently small so that the observation point is approximately equidistant to all points on the dipole; that is, assume $R' \approx R$.

6.28 In free space, the magnetic field is given by

$$\mathbf{H} = \hat{\boldsymbol{\phi}}\,\frac{36}{r}\cos(6\times 10^9 t - kz) \quad \text{(mA/m)}.$$

*(a) Determine $k$.

(b) Determine $\mathbf{E}$.

(c) Determine $\mathbf{J}_d$.

6.29 The magnetic field in a given dielectric medium is given by

$$\mathbf{H} = \hat{\mathbf{y}}6\cos 2z \sin(2\times 10^7 t - 0.1x) \quad \text{(A/m)},$$

where $x$ and $z$ are in meters.

(a) Determine $\mathbf{E}$.

(b) Determine the displacement current density $\mathbf{J}_d$.

(c) Determine the charge density $\rho_v$.



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*6.24 $\varepsilon = 4\varepsilon_0$、$\mu = \mu_0$ 和 $\sigma = 0$ 的介电材料中的磁场由下式给出

$$\mathbf{H}(y,t) = \hat{\mathbf{x}}5\cos(2\pi \times 10^7 t + ky) \quad \text{(A/m)}.$$

求 k 和相关电场 E。

6.25 给定电场

$$\mathbf{E} = \hat{\mathbf{x}}E_0 \sin ay \cos(\omega t - kz),$$

其中 $E_0$、a、ω 和 k 是常数，求 H。

*6.26 短偶极子天线辐射的电场在球坐标中给出：

$$\mathbf{E}(R,\theta;t) = \hat{\boldsymbol{\theta}}\,\frac{2\times 10^{-2}}{R}\sin\theta\,\cos(6\pi \times 10^8 t - 2\pi R) \quad \text{(V/m)}.$$

求 H(R, θ;t)。

6.27 赫兹偶极子是一根短导线，在其长度 l 上承载近似恒定的电流。如果这样的偶极子沿 z 轴放置，其中点位于原点，并且如果流过它的电流为 $i(t) = I_0 \cos \omega t$，则找到以下内容：

(a) 球坐标系中观察点 Q(R, θ, ϕ) 处的延迟矢量势 $\widetilde{A}$(R, θ, ϕ)。

(b) 磁场相量 $\widetilde{H}$(R, θ, ϕ)。

假设 l 足够小，使得观测点与偶极子上的所有点大致等距；也就是说，假设 R′ ≈ R。

6.28 在自由空间中，磁场由下式给出

$$\mathbf{H} = \hat{\boldsymbol{\phi}}\,\frac{36}{r}\cos(6\times 10^9 t - kz) \quad \text{(mA/m)}.$$

*(一确定 k。

(b) 确定E。

(c) 确定 $J_d$。

6.29 给定电介质中的磁场由下式给出

$$\mathbf{H} = \hat{\mathbf{y}}6\cos 2z \sin(2\times 10^7 t - 0.1x) \quad \text{(A/m)},$$

其中 x 和 z 的单位为米。

(a) 确定E。

(b) 确定位移电流密度 $J_d$。

(c) 确定电荷密度 $\rho_v$。